

Green's Theorem

If $M(x, y)$ & $N(x, y)$ are continuous function of x and y having continuous partial derivatives in the region R of xy -plane bounded by a closed curve C , then:-

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

→ Green theorem is useful for evaluating of line integral around a closed curve C .

Example:- A vector field $\vec{F}$ is given by $\vec{F} = \sin y \hat{i} + x(1 + \cos y) \hat{j}$ evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$, where C is a circular path given by

$$x^2 + y^2 = a^2.$$

Solⁿ:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \sin y dx + x(1 + \cos y) dy$$

$$M = \sin y \quad N = x(1 + \cos y)$$

By Green's Theorem

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \iint_R (1 + \cos y - \cos y) dx dy$$

$$= \iint_R dx dy = \pi a^2$$

for Area of $x^2 + y^2 = a^2$

Question 8.1

Evaluate $\oint (x dy - y dx)$ around
the circle $C: x^2 + y^2 = 1$.

Solⁿ —

Question #02 Verify Green's Theorem in the plane for

$$\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy \text{ bounded by } x=0, y=0; x+y \leq 1$$

Solⁿ :—

$$\oint_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

* Green's theorem is—

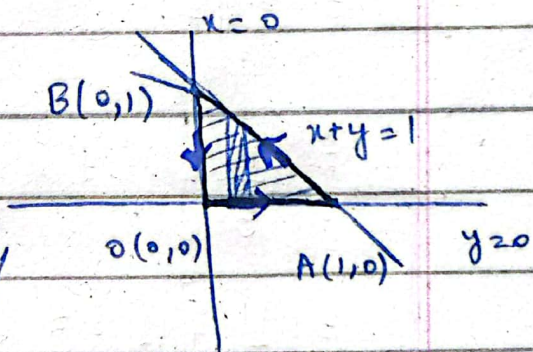
$$\therefore \oint M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} + \frac{\partial M}{\partial y} \right) dy dx$$

L.H.S

$$= \oint M dx + N dy$$

$$= \oint (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

$$= \int_{OA} M dx + N dy + \int_{AB} M dx + N dy + \int_{BO} M dx + N dy$$



$$= I_1 + I_2 + I_3$$

→ By Solving individually

$$\Rightarrow I_1 = \oint_{OA} (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

$$\because y=0 \Rightarrow dy=0$$

$$I_1 = \int_{OA} 3x^2 dx = 3 \int_0^1 x^2 dx$$

$$= 3 \left| \frac{x^3}{3} \right|_0^1 = (1)^3 - (0)^3 \quad \boxed{I_1 = 1}$$

$$\rightarrow P I_2 = \oint_{AB} (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

$x+y=1 \Rightarrow y=0 \Rightarrow x=1$ $x=0$
 $x+y=1 \Rightarrow x=1-y \Rightarrow dx = -dy$

$$I_2 = \int_0^1 [3(1-y)^2 - 8y^2](-dy) + (4y - 6(1-y)y) dy$$

$$= \int_0^1 (3 - 6y + 3y^2 - 8y^2)(-dy) + (4y - 6y + 6y^2) dy$$

$$= \int_0^1 (-3 + 6y + 5y^2) dy + (2y + 6y^2) dy$$

$$= \int_0^1 (-3 + 6y + 5y^2 - 2y + 6y^2) dy$$

$$= \int_0^1 (11y^2 + 4y - 3) dy$$

$$= 11 \left| \frac{y^3}{3} \right|_0^1 + 4 \left| \frac{y^2}{2} \right|_0^1 - 3 \left| y \right|_0^1$$

$$= \frac{11}{3} (1)^3 + 2(1)^2 - 3(1)$$

$$= \frac{11}{3} + 2 - 3 = \frac{11}{3} - 1 = \frac{8}{3} \quad \boxed{I_2 = \frac{8}{3}}$$

$$\rightarrow I_3 = \oint_{BO} (3x^2 - 8y^2) dx + (4y - 6xy) dy$$

$x=0, dx=0$ $y=1 \text{ to } 0$

$$= \int_1^0 4y dy = 4 \left| \frac{y^2}{2} \right|_1^0 = 2(0-1)$$

$$\boxed{I_3 = -2}$$

$$L.H.S = I_1 + I_2 + I_3$$

$$= 1 + \frac{8}{3} - 2 = \frac{5}{3} \quad \checkmark$$

$$R.H.S = \iint_R \left(\frac{\partial M}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$= \iint_R -6y(-16y) dy dx = \iint_R 10y dy dx$$

$$= \int_0^1 \int_0^{x-1} 10y dy dx = \int_0^1 10 \left| \frac{y^2}{2} \right|_0^{x-1} dx$$

$$= 5 \int_0^1 (x-1)^2 dx = 5 \int_0^1 (x^2 - 2x + 1) dx$$

$$= 5 \left[\left| \frac{x^3}{3} \right|_0^1 - 2 \left| \frac{x^2}{2} \right|_0^1 + \left| x \right|_0^1 \right]$$

$$= 5 \left(\frac{1}{3} - 1 + 1 \right) = \frac{5}{3} \quad \checkmark$$

$$L.H.S = R.H.S$$

verified:—

Question #03

Using Green's theorem evaluate for $\vec{F} = x^2 \hat{i} + xy \hat{j}$ over the region R enclosed by $y = x^2$ & the line $y = x$.

Solⁿ—

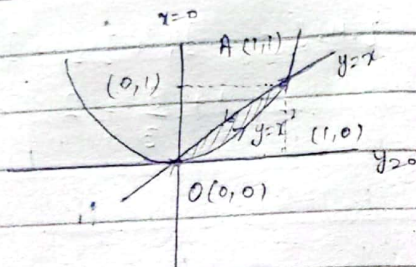
$$y = x^2$$

$$x - x^2 = 0$$

$$x(x-1) = 0$$

$$x=0 \Rightarrow x=1$$

$$y=0 \Rightarrow y=1$$



$$\int \vec{F} \cdot d\vec{r}$$

$$= \int (x^2 \hat{i} + xy \hat{j}) (dx \hat{i} + dy \hat{j})$$

$$= \int x^2 dx + xy dy$$

→ By Green's Theorem:

$$\oint M dx + N dy = \iint \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

L.H.S

$$= \oint M dx + N dy$$

$$= \oint_{OA} M dx + N dy + \oint_{AB} M dx + N dy$$

$$= I_1 + I_2$$

$$I_1 = \int_{OA} x^2 dx + xy dy$$

$$\bullet x=0 \Rightarrow 1$$

$$\bullet y=x^2$$

$$= \int_0^1 x^2 dx + \int_0^1 x \cdot x^2 \cdot 2x dx$$

$$dy = 2x dx$$

$$= \int_0^1 x^2 dx + \int_0^1 x^4 dx = \left[\frac{x^3}{3} \right]_0^1 + \left[\frac{x^5}{5} \right]_0^1$$

$$= \frac{1}{3} + \frac{2}{5} = \frac{1}{3} + \frac{2}{5}$$

$$= \frac{5+6}{15} = \frac{11}{15} \quad \boxed{I_1 = \frac{11}{15}}$$

$$I_2 = \int_{AB} x^2 dx + xy dy$$

$$y(0 \rightarrow 1)$$

$$x=y$$

$$dx=dy$$

$$= \int_0^1 y^2 dy + y^2 dy$$

$$= \int_0^1 2y^2 dy = 2 \left[\frac{y^3}{3} \right]_0^1 = \frac{2}{3} (1-0) = \frac{2}{3}$$

$$I_2 = \frac{2}{3} (-1) = -\frac{2}{3} \quad \boxed{I_2 = -\frac{2}{3}}$$

$$\Rightarrow I_1 + I_2 = \frac{11}{15} - \frac{2}{3} = \frac{11-10}{15}$$

$$L.H.S = \frac{1}{15}$$

$$R.H.S = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$= \iint (y-0) dy dx = \iint_{x^2}^x y dy dx$$

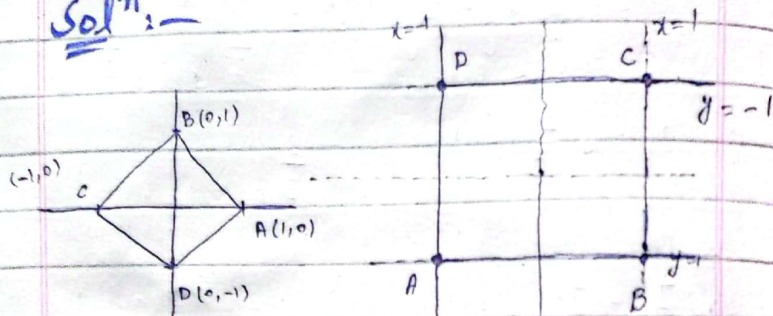
$$= \int_0^1 \left[\frac{y^2}{2} \right]_{x^2}^x dx = \frac{1}{2} \int_0^1 (x^2 - x^4) dx$$

$$= \frac{1}{2} \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right)$$

$$= \frac{1}{2} \left(\frac{2}{15} \right) = \frac{1}{15} \Rightarrow R.H.S$$

L.H.S = R.H.S
verified

Q#4 Verify Green theorem in the plane $\int (x^2+xy)dx + (x^2+y^2)dy$ where C is the square formed by the lines $x = \pm 1, y = \pm 1$.
Soln:



$$\text{L.H.S} = \int_C Mdx + Ndy$$

$$= \int_{AB} Mdx + Ndy + \int_{BC} Mdx + Ndy + \int_{CD} Mdx + Ndy + \int_{DA} Mdx + Ndy$$

$$= I_1 + I_2 + I_3 + I_4$$

$$\Rightarrow I_1 = \int_{-1}^1 (x^2+xy)dx + (x^2+y^2)dy$$

$$I_1 = \int_{-1}^1 (x^2+x)dx = \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_{-1}^1$$

$$= \frac{1}{3}(1+1) + \frac{1}{2}(1-1) = \frac{1}{3}(2)$$

$$\boxed{I_1 = \frac{2}{3}}$$

$$I_2 = \int_{-1}^1 (1+y^2)dy$$

$$\Rightarrow I_2 = \int_{BC} (x^2+xy)dx + (x^2+y^2)dy$$

$$x=1 \Rightarrow dx=0$$

$$= \int_{-1}^1 (1+y^2)dy = \left[y \right]_{-1}^1 + \left[\frac{y^3}{3} \right]_{-1}^1$$

$$= (1-1) + \frac{1}{3}(1-1) = -2 + \frac{1}{3}(-2)$$

$$= \frac{-6-2}{3} = \frac{-8}{3} \quad \boxed{I_2 = \frac{-8}{3}}$$

$$\Rightarrow I_3 = \int_{CD} (x^2+xy)dx + (x^2+y^2)dy$$

$$y=1 \Rightarrow dy=0$$

$$I_3 = \int_{-1}^1 (x^2+x)dx$$

$$I_3 = \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_{-1}^1 = \frac{1}{3}(1-1) + \frac{1}{2}(1-1)$$

$$I_3 = \frac{1}{3}(-2) = \frac{-2}{3} \quad \boxed{I_3 = \frac{-2}{3}}$$

$$\Rightarrow I_4 = \int_{DA} (x^2+xy)dx + (x^2+y^2)dy$$

$$x=-1 \Rightarrow dx=0$$

$$I_4 = \int_{-1}^1 (1+y^2)dy$$

$$= \left[y \right]_{-1}^1 + \left[\frac{y^3}{3} \right]_{-1}^1 = (1-1) + \frac{1}{3}(1-1)$$

$$= 2 + \frac{2}{3} = \frac{6+2}{3} = \frac{8}{3}$$

$$I_4 = \frac{8}{3}$$

$$\begin{aligned} \text{L.H.S} &= I_1 + I_2 + I_3 + I_4 \\ &= \frac{2}{3} - \frac{8}{3} - \frac{2}{3} + \frac{8}{3} = 0 \end{aligned}$$

$$\text{L.H.S} = 0 \quad \checkmark$$

$$\text{R.H.S} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$\begin{aligned} \frac{\partial N}{\partial x} &= \frac{\partial}{\partial x} (x^2 + y^2) = 2x \\ \frac{\partial M}{\partial y} &= \frac{\partial}{\partial y} (x^2 + xy) = x \end{aligned}$$

$$\begin{aligned} \Rightarrow &= \iint_R (2x - x) dy dx \\ &= \int_{x=-1}^1 \int_{y=1}^{-1} x dy dx = \int_{-1}^1 x \left[y \right]_{-1}^1 dx \\ &= \int_{-1}^1 x(-1-1) dx = \int_{-1}^1 -2x dx \\ &= -2 \left[\frac{x^2}{2} \right]_{-1}^1 = -(1-1) = 0 \end{aligned}$$

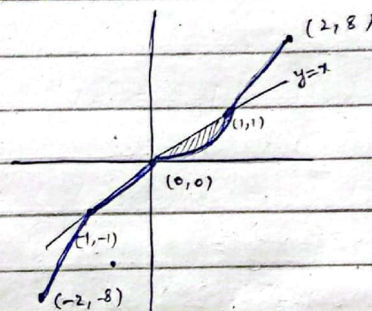
$$\text{R.H.S} = 0 \quad \checkmark$$

$$\text{L.H.S} = \text{R.H.S} \quad \text{verified}$$

Q#05 Using Green's Theorem to evaluate $\int_C x^2 dx + y^2 dy$, where C is the closed path formed by $y=x$ & $y=x^3$ from $(0,0)$ to $(1,1)$.

Soln:-

$y = x^3$
0
1
2
8



Green's Theorem -

$$\oint M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$\text{R.H.S} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (y^2) = 0 \quad \left| \quad \frac{\partial M}{\partial y} = \frac{\partial}{\partial y} (yx^2) = x^2 \right.$$

$$= \int_{x=0}^1 \int_{y=x^3}^x -x^2 dy dx = - \int_0^1 x^2 \left[y \right]_{x^3}^x dx$$

$$= - \int_0^1 x^2 (x - x^3) dx$$

$$= - \int_0^1 (x^3 - x^5) dx = - \left[\frac{x^4}{4} \right]_0^1 + \left[\frac{x^6}{6} \right]_0^1$$

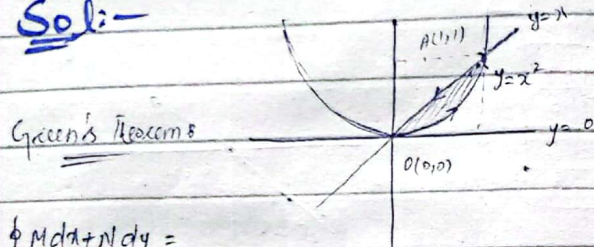
$$= -\frac{1}{4} (1-0) + \frac{1}{6} (1-0) = -\frac{1}{4} + \frac{1}{6}$$

$$= -\frac{3+2}{12} = -\frac{1}{12}$$

$$= -\frac{1}{12}$$

Q#06 :- Verify Green Theorem
for $\oint_C (xy+y^2)dx + x^2dy$
where C is bounded by the curve
 $y=x$ & $y=x^2$.

Sol:-



$$\oint Mdx + Ndy =$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$\text{L.H.S} = \oint Mdx + Ndy$$

$$= \oint_{OA} Mdx + Ndy + \oint_{AO} Mdx + Ndy$$

$$= I_1 + I_2$$

$$\Rightarrow I_1 = \int_{OA} (xy+y^2)dx + x^2dy$$

$y=x^2$
 $dy=2xdx$

$$= \int_0^1 (x \cdot x^2 + x^4)dx + x^2 \cdot 2xdx$$

→ point of intersection.
 $x^2=x$
 $x^2-x=0$
 $x(x-1)=0$
 $x=0, x=1$
Similarly,
 $y=0, y=1$

$$= \int_0^1 (x^3 + x^4 + 2x^3)dx$$

$$= \int_0^1 (3x^3 + x^4)dx$$

$$= 3 \left| \frac{x^4}{4} \right|_0^1 + \left| \frac{x^5}{5} \right|_0^1$$

$$= \frac{3}{4}(1) + \frac{1}{5}(1) = \frac{3}{4} + \frac{1}{5}$$

$$= \frac{15+4}{20} = \frac{19}{20} \Rightarrow I_1 = \frac{19}{20}$$

$$\Rightarrow I_2 = \int_{AO} (xy+y^2)dx + x^2dy$$

x varies from
 $y=x \Rightarrow dy=dx$ 1 to 0

$$= \int_1^0 (x \cdot x + x^2)dx + x^2dx$$

$$= \int_1^0 (x^2 + x^2 + x^2)dx = \int_1^0 3x^2dx$$

$$= 3 \left| \frac{x^3}{3} \right|_1^0 = [0 - 1] = -1 \Rightarrow I_2 = -1$$

$$\therefore \text{L.H.S} = I_1 + I_2$$

$$= \frac{19}{20} - 1 = \frac{19-20}{20} = \frac{-1}{20}$$

$$\text{R.H.S} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x}(x^2) = 2x \quad \left| \quad \frac{\partial M}{\partial y} = \frac{\partial}{\partial y}(xy+y^2) = x+2y \right.$$

$$= \iint_R [2x - (x+2y)] dy dx$$

$$= \int_{x=0}^1 \int_{y=x^2}^x (2x - x - 2y) dy dx$$

$$= \int_0^1 \int_{x^2}^x (x - 2y) dy dx = \int_0^1 \left[x \cdot y \Big|_{x^2}^x - 2 \left| \frac{y^2}{2} \right|_{x^2}^x \right] dx$$

$$= \int_0^1 x(x - x^2) - (x^2 - x^4) dx$$

$$= \int_0^1 (x^2 - x^3 - x^2 + x^4) dx = \int_0^1 (x^4 - x^3) dx$$

$$= \left| \frac{x^5}{5} \right|_0^1 - \left| \frac{x^4}{4} \right|_0^1 = \frac{1}{5} - \frac{1}{4} = \frac{-1}{20}$$

$$= \frac{4-5}{20} = \frac{-1}{20} \quad \checkmark$$

L.H.S = R.H.S Verified.

Q#7 Use Green's Theorem in plane to evaluate $\oint (x+2y)dx + (y+3x)dy$ where C is the circle $x^2+y^2=1$.

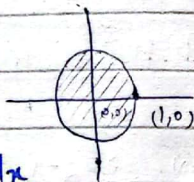
Solⁿ:

$$\oint (x+2y)dx + (y+3x)dy$$

$\Rightarrow$ Green's Theorem:-

$$\oint Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$R.H.S = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$



$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (y+3x) = 3 \quad \left| \quad \frac{\partial M}{\partial y} = \frac{\partial}{\partial y} (x+2y) = 2 \right.$$

$$\Rightarrow R.H.S = \iint (3-2) dy dx = \iint 1 dy dx$$

$\rightarrow$ This is the area of circle:-

that is equal to (πr^2)

$$= \pi r^2 \quad \because (r=1)$$

$$= \pi \quad \boxed{= \pi}$$

Q#08:- Verify Green's Theorem

for the field $\vec{F} = 2x\hat{i} - 3y\hat{j}$. Disc

forms the region $x^2+y^2 \leq a^2$ &

its boundary $\vec{r}(t) = a \cos t \hat{i} + a \sin t \hat{j}$

Solⁿ:

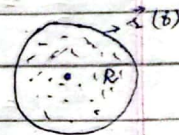
$$\vec{F} = 2x\hat{i} - 3y\hat{j}$$

$$\vec{r}(t) = a \cos t \hat{i} + a \sin t \hat{j}$$

$$M = 2x, \quad N = -3y$$

$$x = a \cos t, \quad y = a \sin t$$

$$dx = -a \sin t dt, \quad dy = a \cos t dt$$



$\Rightarrow$ Green's Theorem:

$$\oint Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$L.H.S = \oint Mdx + Ndy$$

$$= \oint (2x)(-a \sin t dt) + (-3y)(a \cos t dt)$$

$$= \int 2(a \cos t)(-a \sin t dt) - 3(a \sin t)(a \cos t dt)$$

$$= \int -2a^2 \sin t \cos t dt - 3a^2 \sin t \cos t dt$$

$$= a^2 \int -5 \sin t \cos t dt$$

$$= -\frac{5a^2}{2} \int 2 \sin t \cos t dt$$

$$= -\frac{5a^2}{2} \int_0^{2\pi} \sin 2t dt$$

$$= -\frac{5a^2}{2} \left[-\frac{\cos 2t}{2} \right]_0^{2\pi}$$

$$= -\frac{5a^2}{4} (\cos 4\pi - \cos 0)$$

$$\begin{aligned} x^2 + y^2 &= r^2 \\ x &= r \cos \theta \\ y &= r \sin \theta \\ 0 &\leq \theta \leq 2\pi \end{aligned}$$

$$\therefore \cos n\pi = (-1)^n$$

$$\cos 4\pi = 1, \cos 0 = 1$$

$$= \frac{5a^2}{4} (1-1) = 0 \quad \checkmark$$

= R.H.S

$$\text{R.H.S} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (3y) = 0 \quad \frac{\partial M}{\partial y} = \frac{\partial}{\partial y} (2x) = 0$$

$$= \iint (0-0) dy dx$$

$$\boxed{\text{R.H.S} = 0} \quad \checkmark$$

L.H.S = R.H.S
Verified

Q#09:-

Verify Green's theorem in XY-plane for $\int x^2 dx + xy dy$ where C is the boundary of the region given by $x=0, y=0, x=a, y=a$ ($a>0$)

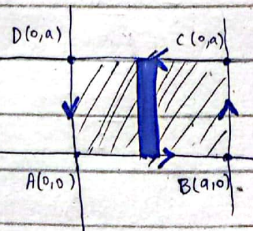
Sol:-

Green's theorem

$$\oint M dx + N dy =$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$= \int_C x^2 dx + xy dy$$



$$\text{L.H.S} = \oint M dx + N dy$$

$$= \int_{AB} M dx + N dy + \int_{BC} M dx + N dy + \int_{CD} M dx + N dy + \int_{DA} M dx + N dy$$

$$\text{L.H.S} = I_1 + I_2 + I_3 + I_4$$

$$\Rightarrow I_1 = \int_{AB} x^2 dx + xy dy = \int_0^a x^2 dx + xy dy$$

$y=0 \Rightarrow dy=0$

$$= \int_0^a x^2 dx = \left| \frac{x^3}{3} \right|_0^a = \frac{a^3}{3} \quad \checkmark$$

$$\Rightarrow I_2 = \int_{BC} x^2 dx + xy dy \quad \begin{matrix} x=a \\ dx=0 \end{matrix}$$

$$= \int_0^a ay dy = a \left| \frac{y^2}{2} \right|_0^a = \frac{a^3}{2} \quad \checkmark$$

$$\Rightarrow I_3 = \int_{CD} x^2 dx + xy dy \quad \begin{matrix} y=a \\ dy=0 \end{matrix}$$

$$= \int_a^0 x^2 dx = \left| \frac{x^3}{3} \right|_a^0 = -\frac{a^3}{3} \quad \checkmark$$

$$\Rightarrow I_4 = \int_{DA} x^2 dx + xy dy \quad \begin{matrix} x=0 \\ dx=0 \end{matrix}$$

$$= \int_a^0 0 = 0 \quad \checkmark$$

$$\Rightarrow \text{L.H.S} = I_1 + I_2 + I_3 + I_4$$

$$= \frac{a^3}{3} + \frac{a^3}{2} - \frac{a^3}{3} + 0 = \frac{a^3}{2} \quad \boxed{\frac{a^3}{2}}$$

$$R.H.S = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy dx$$

$$\bullet \frac{\partial N}{\partial x} = \frac{\partial}{\partial x} (xy) = y \quad \bullet \frac{\partial M}{\partial y} = \frac{\partial}{\partial y} (x^2) = 0$$

$$R.H.S = \int_{x=0}^{x=a} \int_{y=0}^{y=a} y dy dx$$

$$= \int_0^a \left[\frac{y^2}{2} \right]_0^a dx = \int_0^a \frac{a^2}{2} dx$$

$$= \frac{a^2}{2} \left[x \right]_0^a = \frac{a^3}{2}$$

L.H.S = R.H.S
Verified.

Q#10

Evaluate by Green's theorem

$\int_C (x^2 - \cos hy) dx + (y + \sin x) dy$
where C is the rectangle in the vertices $(0,0)$, $(\pi,0)$, $(\pi,1)$, $(0,1)$.

Sol:-

$$\int_C (x^2 - \cos hy) dx + (y + \sin x) dy$$